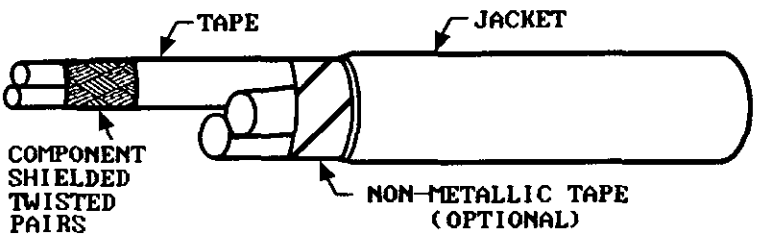
INCH-POUND
MIL-C-24640/16B
31 July 1989
SUPERSEDING
MIL-C-0024640/16A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/16
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, AWG 18, TYPE 2XSW (INCLUDING VARIATION TYPES 2XSOW AND 2XSWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.



SH 12763

COMPONENT SHIELDED PAIR:

(B) First - Tin coated copper conductor, tin coated, AWG 18, 7 stranding in accordance with ASTM B 286. Nominal diameter of stranded conductor 0.049 inch.

Second - Material requirements in accordance with MIL-W-81044/12 except the conductor and identification codes and methods shall be in (B) accordance with MIL-C-24640. Twisted pair identification code applied by method 2 over shields and method 6 on component wires. Finished wire diameter 0.062 inch minimum and 0.068 inch maximum.

Third - Shield shall be tin-coated copper braid with a coverage not (B) less than 90 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (watertight).

(B) denotes changes.

AMSC N/A FSC 6145

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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(B) TABLE I. Construction details-unarmored.

PIN M24640/16	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01UN	2XSW-1	1	0.034	0.240	0.258	52.9
-02UN	2XSW-3	3	.041	.436	.470	167.0
-03UN	2XSW-7	7	.044	.573	.617	318.0

(B) TABLE II. Construction details-armored.

PIN M24640/16	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01AN	2XSWA-1	1	0.034	0.290	0.308	68.9
-02AN	2XSWA-3	3	.041	.486	.520	210.4
-03AN	2XSWA-7	7	.044	.623	.667	375.2

(B) TABLE III. Construction details-overall shield.

PIN M24640/16	Type number	Number of pairs	Diameter over shield (inches) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inches)		Cable weight (pounds per 1000 feet) (max)
					(min)	(max)	
-01U0	2XSOW-3	3	0.374	0.041	0.487	0.525	227.0
-02U0	2XSOW-7	7	.514	.044	.608	.656	401.0
-03U0	2XSOW-12	12	.702	.047	.802	.864	626.0
-04U0	2XSOW-19	19	.844	.050	.938	1.010	884.0
-05U0	2XSOW-30	30	1.080	.057	1.180	1.270	1378.0

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REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

(B) Finished wire:

For the finished wire the following additional tests shall be applicable in accordance with MIL-W-81044 :

WIRE RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 135°C (275°F) maximum conductor temperature.
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 250 ± 2°C (482 ± 3.6°F) for 6 hours; identification legibility, 225 ± 2°C (437 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F).
Color: In accordance with MIL-STD-104, class 1; white preferred.
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Identification, striping, or banding durability: 125 cycles (250 strokes) (min), 500-gram weight.
Impulse dielectric test:
Primary insulation (when test is used in lieu of spark test): 6.0 kilovolts (peak), 100 percent test.
Finished wire: 8.0 kilovolts (peak), 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 120 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms-inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

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When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) Braided metal armor shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(a) An overall tin-coated copper braid shall be added.
(B) (b) Braid coverage shall be not less than 90 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum (B)	7.11
Voltage withstand - volts, root mean square, minimum (B)	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield (type 2XSOW only)	500
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity (type 2XSOW only)	No failure
Jacket flaws (type 2XSOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum (B)	60
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms (B)	30 min, 53 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	4
Gas flame (1 hour)	Not applicable

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EXAMINATION AND TESTS: - Continued

	Requirements
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type 2XSWA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance (type 2XSOW only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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QUALIFICATION INSPECTION: - Continued

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-1004-16)

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR
Navy - CG

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